

Project Design Phase

Solution Architecture

Date	
Team ID	
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process — with many sub-processes — that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best technology solution to solve the existing network change request problem.
- Describe the structure, characteristics, behavior, and other aspects of the ServiceNow solution to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Proposed Architecture Overview

The solution is architected natively on the ServiceNow platform, spanning four functional layers: a self-service intake layer, a workflow and validation layer, an orchestration and fulfillment layer, and an audit / reporting layer. Together these layers automate the full lifecycle of a network change request — from submission to audit-ready closure — while keeping the CMDB continuously synchronized with the real network state.

Solution Architecture Diagram:

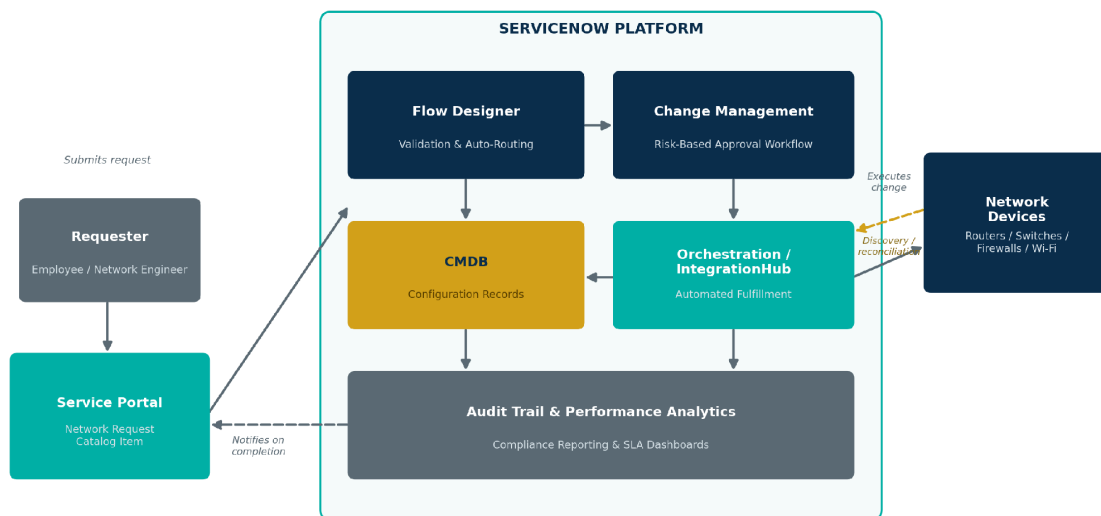


Figure 1: Solution architecture and data flow of the Automated Network Request Management application

Architecture Walkthrough

1. Requesters submit a network change request through the ServiceNow Service Portal using a guided Network Request catalog item.
2. Flow Designer automatically validates the request (mandatory fields, duplicate check, CMDB correlation) and classifies it by network domain.
3. The validated request is routed into ServiceNow Change Management, which drives a risk-based, automated approval workflow with the appropriate approvers or CAB.
4. Once approved, Orchestration / IntegrationHub triggers automated fulfillment, executing the configuration change on the relevant network devices.
5. The CMDB is automatically updated with the new configuration state, and discovery/reconciliation confirms the change matches the live network.
6. An audit-ready change record is generated automatically, and the requester is notified of completion; performance analytics dashboards capture SLA and compliance metrics.